Buildings
Problem ID: buildings

As a traveling salesman in a globalized world, Alan has always moved a lot. He almost never lived in the same town for more than a few years until his heart yearned for a different place. However, this newest town is his favorite yet - it is just so colorful. Alan has recently moved to Colorville, a smallish city in between some really nice mountains. Here, Alan has finally decided to settle down and build himself a home - a nice big house to call his own. In Colorville, many people have their own houses - each painted with a distinct pattern of colors such that no two houses are the same. Every wall consists of exactly $n \times n$ bricks, each painted with a separate color. The walls of the houses are arranged in the shape of a regular m-gon, with a roof on top.

Of course, Alan wants to follow this custom to make sure he fits right in. However, there are just so many possible designs to choose from. Can you tell Alan how many possible house designs there are? (Two house designs are obviously the same if they can be translated into each other just by rotation)

Input

The input consists of one line with three integers n ($1 \leq n \leq 500$), m ($3 \leq m \leq 500$) and c ($1 \leq c \leq 500$), where Alans house should have the shape of a regular m-gon with every wall consisting of $n \times n$ bricks with each brick colored in one of c possible colors.

Output

Output s where s is the number of possible different house designs. Since s can be very large, output $s \mod 1000000007$.

Sample Input 1	Sample Output 1
1 3 1	1

Sample Input 2	Sample Output 2
2 5 2	209728